Geospatial Contextual Prior-Enabled Knowledge Reasoning Framework for Fine-Grained Aircraft Detection in Panoramic SAR Imagery

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***Abstract*— Fine-grained aircraft detection from synthetic aperture radar (SAR) imagery is of significance in transportation and military domains. Based on the prior knowledge that aircraft are frequently found in airports, current research adopts airport-to-aircraft detection pipelines for aircraft detection and classification in panoramic SAR images. Geospatial information, including the approximate airport location and the spatial relationships between the airport and aircraft, represents valuable supplementary information that can enhance fine-grained aircraft detection performance. However, due to unreliable geographical information and the limited perceptual field, it is challenging to detect and classify aircraft in panoramic SAR imagery. To address this, a novel geospatial contextual prior-enabled knowledge reasoning framework is proposed. First, an unsupervised and lightweight geospatial-driven airport detection (AD) method is presented by combining geographic information matching and optical-to-SAR image registration, which can quickly locate airports and narrow the scope for fine detection. Then, an improved real-time model for object detection with a recursive-gated spatial interaction module (RTMDet-RSIM) is proposed for fine-grained aircraft detection. RTMDet-RSIM uses frequency-domain analysis to improve global information modeling ability without excessive computational cost. Finally, a relational prior-based reasoning strategy is proposed by modeling the geospatial category relationships from historical images to strengthen classification performance. Experiments on 61 panoramic SAR images covering 13 aircraft categories show that the proposed method achieves high detection accuracy with a mean average precision (mAP) of 81.2%, while the average test time for an image size of 8738 × 7636 is about 8.58 s. The source code and dataset will be released.**

***Index Terms*— Deep learning, fine-grained aircraft detection, geospatial category relationships, global information, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR), known for its exceptional prowess in acquiring high-resolution imagery across diverse weather conditions, has been widely used in a variety of fields [1]. Aircraft is a valuable transportation means and combat resource. Therefore, the accurate detection and classification of aircraft play a vital role in military situational awareness and civil airport management [2].

In practical applications, SAR images are often presented to users in a panoramic view. Aircraft detection and classification in very large coverage face two challenges: low efficiency and high false alarms. Based on the prior knowledge that aircraft are frequently found within airports, airport detection (AD) is used to reduce processing scope efficiently, enhancing the efficiency of aircraft detection and classification [3], [4], [5]. Therefore, the existing aircraft detection and classification framework consists of three main stages: AD, aircraft detection and classification, and redundant object elimination, as shown in Fig. 1(a). Specifically, the first and third steps can also be seen as the preprocessing and postprocessing steps of aircraft detection and classification. Furthermore, aircraft detection and classification accuracy mainly depends on the geospatial positioning reliability and the aircraft type separation capability, where the former refers to the first and third stages, and the latter refers to the second stage. We will review the existing methods from these two aspects.

For the aspect of geospatial positioning reliability, geospatial information is valuable additional information that can be used to constrain the scope of fine-grained aircraft detection. On one hand, airports and ports, as fixed infrastructures with clear geographical boundaries, offer predefined geolocation priors, such as airport coordinates and coastline boundaries. These priors can help quickly locate the region of interest, significantly reducing the need for extensive labeling and computational resources [6], [7], [8]. However, due to the inherent position errors in the imaging process [9], [10], [11], the direct application of geographic information matching may lead to inaccurate AD, which in turn hinders the performance of fine-grained aircraft detection in panoramic SAR imagery. On the other hand, in the planning and design of airport aprons, different aircraft types are typically distributed across designated areas to optimize operational efficacy. The partitioned spatial arrangement remains consistent across images from different periods. Based on this knowledge, humans can

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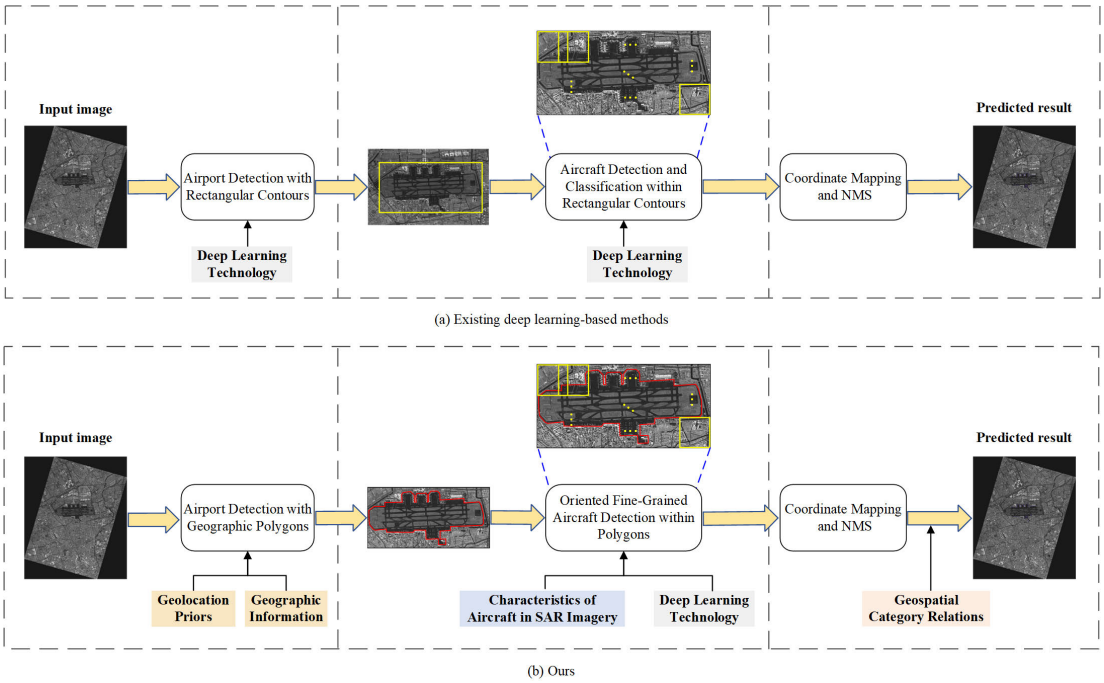


Fig. 1. Schematic of aircraft detection and classification framework. (a) Workflow of existing deep-learning-based methods and (b) workflow of our proposed framework.

analyze historical images to make logical inferences about the aircraft category and location distributions. This discernment, rooted in analyzing the geospatial category relationships of aircraft at airports from historical images, can emerge as a supplement to alleviate prediction bias in detection networks and enhance the performance of fine-grained aircraft detection but has not been studied yet. The main challenges in modeling and exploiting geospatial categorical relationships are reflected in two aspects: how to associate historical images from different time periods to model geospatial categorical relationships, and how to effectively integrate geospatial categorical relationships with deep learning models for high-precision predictions.

For the aspect of aircraft type separation, fine-grained aircraft detection aims to locate and classify specific aircraft types simultaneously [12]. Due to the coherent imaging mechanisms of SAR, aircraft often appear as discrete scatter points with blurred shapes, making aircraft detection and classification more challenging than optical images. To tackle this challenge, some scholars have proposed to design efficient multiscale feature fusion modules and attention mechanisms to capture essential information of aircraft across different scales [13], [14], [15], [16], [17]. Others have explored using geometric structure [18], scattering topology [19], and polarimetric scattering characteristics [20], [21], [22] to mitigate incomplete detection caused by the aircraft's discrete appearance in SAR imagery. Furthermore, contrastive learning has been used to enhance class separability and improve the performance of fine-grained aircraft detection [12]. However, the above-mentioned methods mainly extract deep features of SAR images through the convolutional neural networks (CNNs) for aircraft detection and classification, and the limited perceptual field prevents them from effectively capturing global information. Global information can provide valuable insights into the relationships between discrete scattering points, guiding the network to better capture discriminative features among fine-grained subcategories [23].

To bridge these gap, a more systematic and promising framework is provided in this article to heighten the efficiency and accuracy of aircraft detection and classification, as shown in Fig. 1(b). First, leveraging geolocation priors, geodatabase, and image registration technologies are introduced to improve the accuracy of AD. Meanwhile, the proposed method is an unsupervised and lightweight method which can perform high-precision AD and provide guidance for fine-grained aircraft detection in panoramic SAR imagery. Then, an improved real-time model for object detector with a recursive-gated spatial interaction module (RTMDet-RSIM) is presented. In RTMDet-RSIM, the recursive gated convolution [24] and frequency-domain analysis are used to improve CNN's global information modeling ability without excessive computational cost, thereby enhancing the performance of fine-grained aircraft detection. Finally, a relational prior-based global reasoning strategy is proposed. Specifically, the geospatial category probability maps are modeled using Gaussian distributions (GDs), which effectively capture the inherent uncertainty and variability associated with the geospatial category relationships. On this basis, a postprocessing model that collaborates with the detection network is developed to reduce false positives.

The main contributions in this article can be summarized as follows.

1) An innovative geospatial contextual prior-enabled knowledge reasoning framework is crafted by making reasonable use of geolocation priors, geospatial category relationships, and the aircraft's scattering characteristics

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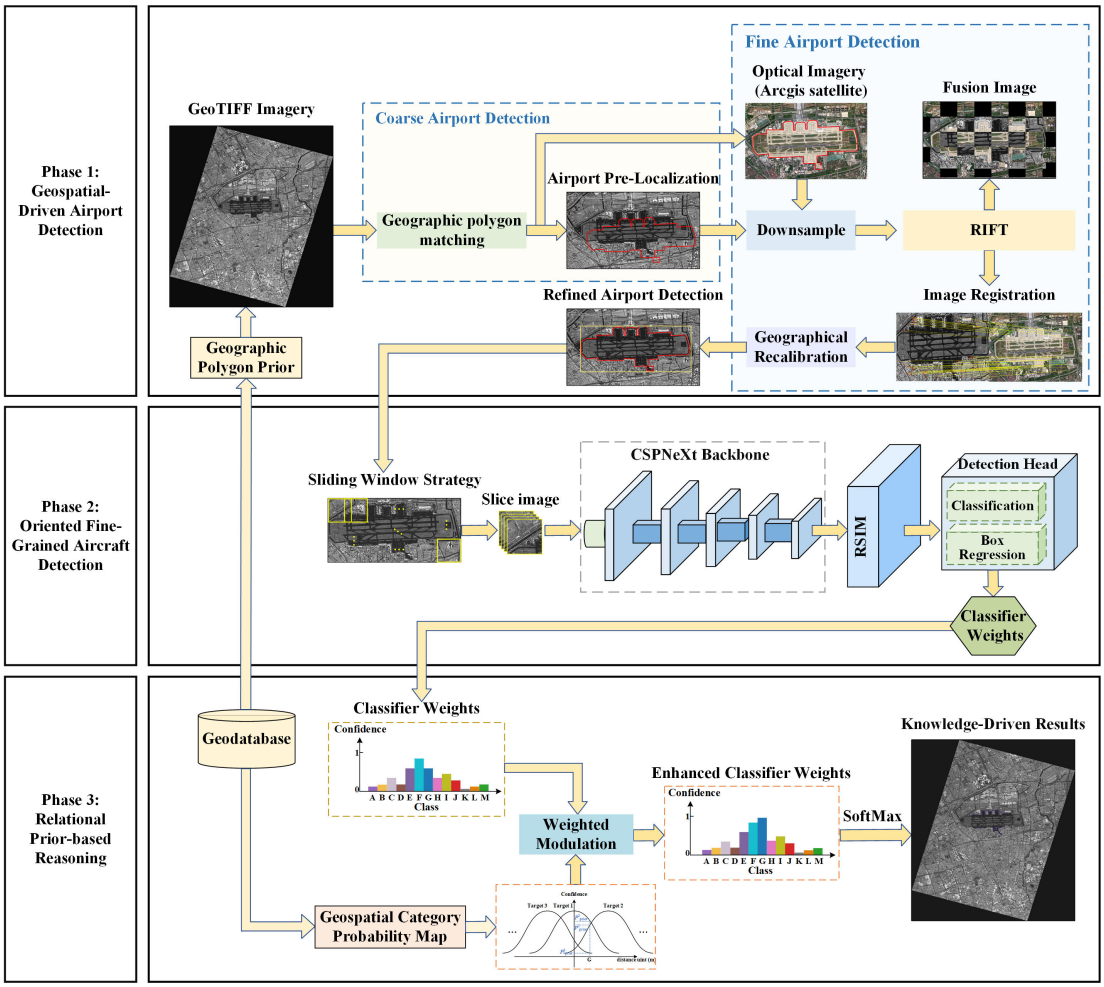


Fig. 2. Overall architecture of the proposed framework. It involves three phases. Phase 1—geospatial-driven AD, phase 2—oriented fine-grained aircraft detection, and phase 3—relational prior-based reasoning aims to achieve efficient fine-grained aircraft detection in panoramic SAR imagery.

for efficient fine-grained aircraft detection from complex panoramic SAR imagery.

2) We integrate geospatial information into AD and postprocessing stages to enhance fine-grained aircraft detection performance. On one hand, geographic information matching and optical-to-SAR image registration are used to rapidly locate airports and narrow the search area for fine detection. On the other hand, a relational prior-based reasoning strategy is designed using geospatial category relationships to complement the DL-based model and minimize category confusion.
3) An oriented fine-grained aircraft detection network, RTMDet-RSIM, is proposed. Benefiting from frequency-domain analysis and a recursive gating mechanism, RTMDet-RSIM enables efficient global information modeling to better encode the relationships among aircraft's backscattering features, thus enhancing the performance of fine-grained aircraft detection.

The remainder of this article is organized as follows. Section II provides the proposed method in detail. In Section III, the experimental results and analysis are presented. Finally, the conclusion and future research are discussed in Section IV.

II. Methodology

The overview architecture of the proposed framework is shown in Fig. 2, which consists of three main phases: 1) geospatial-driven AD; 2) oriented fine-grained aircraft detection; and 3) relational prior-based reasoning. This section begins with an overview of the proposed framework, followed by each phase described in detail.

A. Overall Architecture

As depicted in Fig. 2, the geospatial-driven AD serves as the initial phase, where panoramic SAR images are received as input. Geographic information extracted from SAR images is seamlessly matched with geographic polygons stored in the database, enabling rapid AD. To compensate for inevitable positioning inaccuracies during imaging, the radiation-variation insensitive feature transform (RIFT) [25] is implemented by leveraging optical-to-SAR image registration to mitigate geographic information bias, thereby improving the accuracy of AD. Following the acquisition of the airport region, the meticulously designed oriented fine-grained aircraft detection network is carried out as the second phase to obtain preliminary location and category predictions. In the third phase, geospatial category probability maps are constructed

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(a) (b)

Fig. 3. Visualization of AD based on different types of bounding box. (a) Horizontal bounding box overlaid on optical imagery. (b) Geographic polygon produced on the BigeMap platform overlaid on optical imagery.

using ground truths to organize the geospatial relationships between aircraft's categories and their parking positions at the airport. On this basis, a relational prior-based reasoning strategy is devised to rectify category prediction biases, enhancing the performance of fine-grained aircraft detection. Finally, knowledge-driven results are obtained from panoramic SAR images.

B. Geospatial-Driven AD

With time, airports have developed consistent and stable basic structures. Leveraging this knowledge, we create geographic polygons using the BigeMap platform [26], open-source satellite imagery, and geolocation information for the airport region of interest. These geographic polygons and the corresponding optical images are stored in the geographic information database as geographical priors. As shown in Fig. 3, the geographic polygon, which captures the geometric contour of the airport more accurately than the traditional horizontal box, further reduces irrelevant background information. However, due to the inherent position bias in the imaging process, directly matching geospatial information for AD may lead to inaccurate position predictions and the neglect of aircraft within the airport. To address this obstacle, we propose a geospatial-driven AD method with a coarse-to-fine detection process. First, geographic polygon matching is performed based on a geographic database to quickly locate the coarse airport region and associate the corresponding optical images. Subsequently, the RIFT algorithm is used to recalibrate geographic information for SAR imagery. In RIFT, feature matching is conducted between the optical and SAR images, and the corresponding points are identified. Concurrently, image downsampling is used to optimize the efficiency of the algorithm. Then, geographical transformation parameters are calculated for SAR images by combining the geographical information of the optical image with the extracted corresponding points. As a result, refined geographic information can be obtained. This process facilitates more precise AD and greatly boosts the efficiency of fine-grained aircraft detection. The workflow of the proposed algorithm is illustrated in Fig. 4.

C. Oriented Fine-Grained Aircraft Detection

For efficient fine-grained aircraft detection in SAR imagery, we present a novel oriented fine-grained aircraft detection network, named RTMDet-RSIM. RTMDet-RSIM is developed

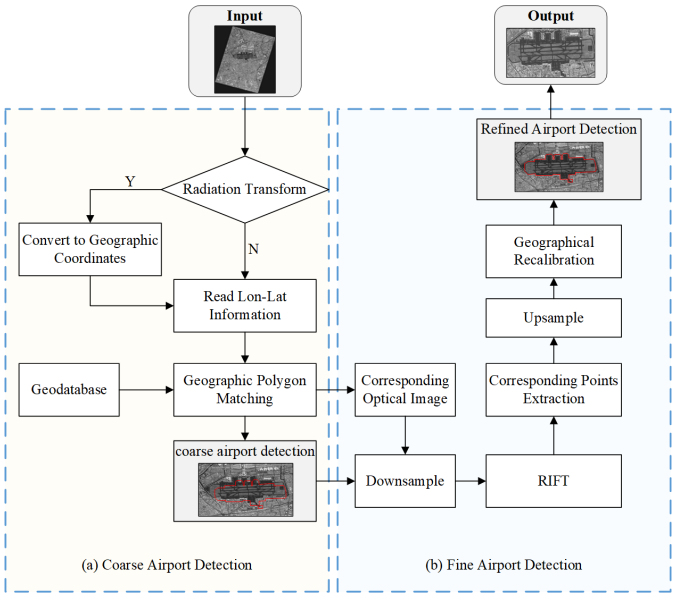


Fig. 4. Workflow of the proposed geospatial-driven AD algorithm. (a) Coarse AD stage using geographic polygon matching. (b) Fine AD stage based on geographic information matching and RIFT.

based on real-time models for object detection (RTMDet) [27] combined with the characteristics of aircraft. As shown in Fig. 5, for the sake of clarity, we propose that its main components consist of feature extraction, classification and regression, and parameter optimization.

First, the CSPNeXt backbone of the RTMDet is tasked with extracting multilevel feature maps $\boldsymbol{P}_0 \sim \boldsymbol{P}_4$ from input images. Subsequently, the proposed recursive-gated spatial interaction module (RSIM) is used to process three smaller feature maps $\boldsymbol{P}_2 \in \mathbb{R}^{128\times128\times192}$, $\boldsymbol{P}_3 \in \mathbb{R}^{64\times64\times384}$, and $\boldsymbol{P}_4 \in \mathbb{R}^{32\times32\times768}$, to capture a more comprehensive feature representation of aircraft through input-adaptive interaction modeling. In this article, we denote the tensor as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times C}$, with H, W, and C representing the height, width, and channel of the feature map, respectively. Finally, $\boldsymbol{P}_{o2} \sim \boldsymbol{P}_{o4}$ are fed into the SepBN head for classification and regression to obtain the oriented bounding boxes (OBBs) and category predictions. The network is optimized by minimizing the loss. In the testing phase, the rotated nonmaximum suppression (rotated NMS) iteratively removes the overlapping bounding boxes to obtain predictions.

1) Real-Time Models for Object Detection: RTMDet is an object detection network with an architecture that prioritizes efficiency and compatibility across both the backbone and neck. Inspired by large-kernel convolution [28], RTMDet introduces 5×5 depthwise convolution and 1×1 pointwise convolution into the basic building blocks and neck in the YOLOX [29] to increase the effective receptive fields for more comprehensive global information modeling. To maintain the tradeoff between speed and accuracy, on one hand, the depth and width of the model are carefully designed which significantly bolsters the model's capability to recognize and localize objects. On the other hand, the SepBN head is adapted to produce the bounding boxes and category predictions via classification and regression. The internal structure of the

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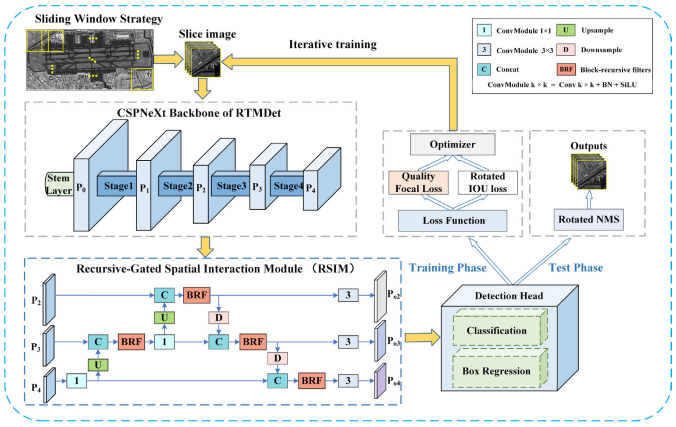


Fig. 5. Structure of the proposed oriented fine-grained aircraft detection network. The input images are passed through RTMDet's CSPNeXt backbone and RSIM for feature extraction and then fed into the detection head for classification and regression. The network is optimized by minimizing the loss and rotated NMS removes overlapping bounding boxes during the testing phase to produce the final predictions.

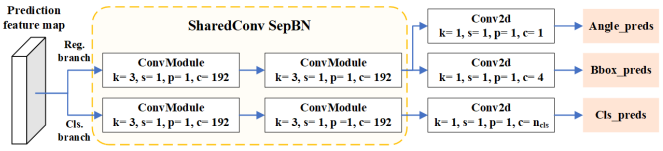


Fig. 6. Structure of SepBN head.

SepBN head is shown in Fig. 6. The 3 × 3 convolution layers facilitate weight parameter sharing and calculate batch normalization (BN) separately across different feature maps, reducing parameters and computational complexity. Following this, three parallel 1 × 1 convolution layers are implemented to predict the location, angle, and category confidence, respectively.

RTMDet is used in this article for its commendable accuracy and speed. However, the complexities inherent in SAR imagery, such as the discrete appearance and scale heterogeneity of aircraft, present formidable obstacles to efficient feature extraction. These limitations exacerbate missed detections and false positives when using RTMDet for fine-grained aircraft detection tasks in SAR imagery, underscoring the urgent need for additional efforts to improve its accuracy.

2) Recursive-Gated Spatial Interaction Module: Given the discrete nature and scale heterogeneity of aircraft, multiscale spatial information interaction is crucial for fine-grained aircraft detection from SAR imagery. Although the self-attention layer can effectively model the spatial interactions within visual data, its computational demands hinder widespread implementation [30]. To overcome this obstacle, the RSIM is proposed in this article. As shown in Fig. 5, the RSIM showcases the bidirectional feature fusion strategy that effectively establishes the complementary relationships among low-level detail and high-level semantic information. The block-recursive filters (BRFs), as the core component of RSIM, exploit global information modeling and efficient input-adaptive interactions to encode the relationship between discrete scatter features of the aircraft, and then provide robust knowledge for the detection head.

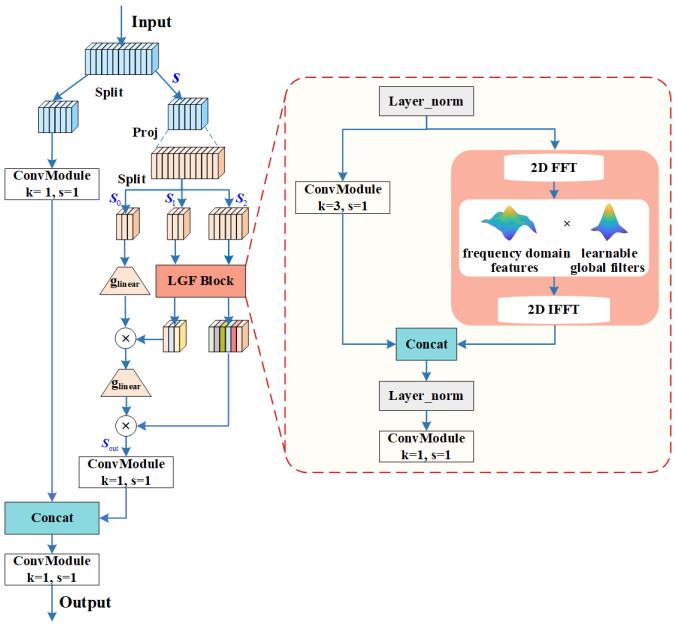


Fig. 7. Structure of BRFs.

The structure of BRFs is shown in Fig. 7. Inspired by the cross stage partial network (CSPNet) [31], the input feature maps are divided along the channel dimension into two distinct branches and operated independently. Subsequently, the features extracted from diverse branches are effectively fused, enhancing feature representation while reducing the floating-point operations (FLOPs). In one branch, the local and global information is progressively extracted and integrated through linear projections and local–global filters (LGFs) blocks to attain more high-level feature representations. Let the input feature $\boldsymbol{X}_{\text{in}}$ and output feature $\boldsymbol{X}_{\text{out}}$ of the BRF module be expressed as follows:

$$\tilde{\boldsymbol{X}}^{H\times W\times C/2}, \boldsymbol{S}^{H\times W\times C/2} = \text{split}\left(\boldsymbol{X}_{\text{in}} \in \mathbb{R}^{H\times W\times C}\right) \tag{1}$$

$$\boldsymbol{X}_{\text{proj}} = \phi_{\text{proj}}(\boldsymbol{S}),\ \boldsymbol{X}_{\text{proj}} \in \mathbb{R}^{H\times W\times C} \tag{2}$$

$$\left[\boldsymbol{S}_0^{H\times W\times C/4}, \boldsymbol{S}_1^{H\times W\times C/4}, \boldsymbol{S}_2^{H\times W\times C/2}\right] = \text{split}\left(\boldsymbol{X}_{\text{proj}}\right) \tag{3}$$

$$\boldsymbol{S}_{\text{out}} = L(\boldsymbol{S}_2) \odot g(L(\boldsymbol{S}_1) \odot g(\boldsymbol{S}_0)),\ \boldsymbol{S}_{\text{out}} \in \mathbb{R}^{H\times W\times C/2} \tag{4}$$

$$\boldsymbol{X}_{\text{out}} = f_{1\times 1}\left(\text{concat}\left(f_{1\times 1}\left(\tilde{\boldsymbol{X}}\right), f_{1\times 1}(\boldsymbol{S}_{\text{out}})\right)\right) \tag{5}$$

where $\phi_{\text{proj}}(\cdot)$ is the projection function, and $\boldsymbol{S}_k$ (k = 0,1,2) is a set of projected features derived from the input feature. $L(\cdot)$ and $g(\cdot)$ represent the LGFs block and dimension matching function, respectively. $\odot$ is the Hadamard product. $f_{1\times 1}(\cdot)$ and concat$(\cdot)$ are 1 × 1 convolution and concatenation operator, respectively.

The LGF block integrates local and global perspectives to enhance the network's ability to capture fine-grained features from SAR images. As illustrated in Fig. 7(a), in the global branch, spatial features are transformed into the frequency domain via a 2-D fast Fourier transform (FFT). Subsequently, trainable global filters [32] are used to learn features in the frequency domain. This methodology is akin to performing convolution in the spatial domain with global kernel size and circular padding, aiming to establish global receptive fields and thoroughly capture feature distributions and patterns.

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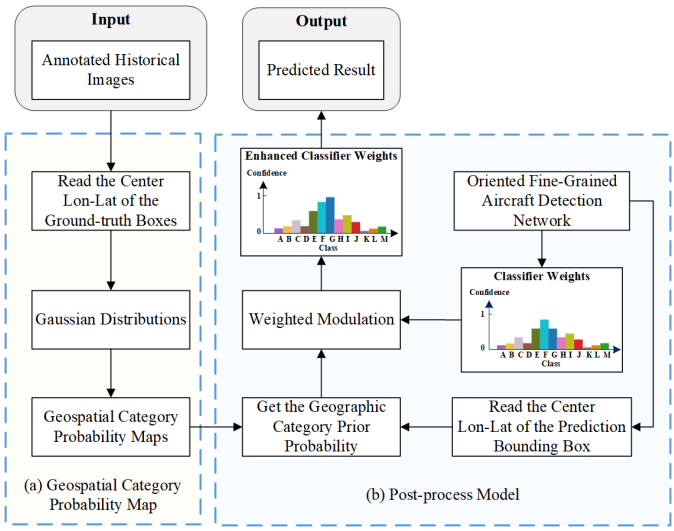


Fig. 8. Workflow of relational prior-based reasoning. (a) and (b) Construction of the geospatial category probability map and postprocess model, respectively.

Furthermore, these processed features are converted back into the spatial domain through a 2-D inverse FFT (IFFT) to maintain spatial consistency and feature integrity. In the local branch, a 3×3 depthwise convolution is used to focus on local features. Finally, the local and global information is processed by a 1×1 convolution and concatenation to heighten the ability of the network to capture significant integral information about aircraft.

D. Relational Prior-Based Reasoning

Due to the discrete appearance and attitude sensitivity of aircraft in SAR imagery, potential aircraft targets may exhibit low confidence levels or closely resemble background clutter or other similar aircraft, which brings ambiguity to the classification. To overcome the obstacle, a relational prior-based reasoning strategy is proposed, which organizes geospatial category relationships and integrates them into the classification process to enhance the performance of the CNN, as shown in Fig. 8. Relational prior-based reasoning comprises two main components: geospatial category probability map and postprocess model. In this section, we first provide details on how a geospatial category probability map is constructed. Then, we elaborate on how category priors, obtained from the geospatial category relationships, are integrated into the classification process.

1) Geospatial Category Probability Map: Aircraft are typically stationed at specific positions and orientations at the airport, with diverse aircraft types allocated to designated parking areas to optimize operational efficacy. Therefore, the probability of encountering a specific aircraft category at a given location can be reasonably inferred. In this article, the probability maps are used to represent the geospatial category relationships of various aircraft types.

During the creation of geospatial category probability maps, the determination of the coordinate system and map resolution is crucial. The longitude and latitude coordinate system is used in this article, which is more in line with the expression of geographic information. The resolution of longitude and latitude determines the spacing between each grid cell within the longitude–latitude coordinate systems. The map resolution defines the actual surface distance covered by each grid cell on the map. These elements are of great importance for defining the granularity of the grid cells in the geospatial map. In this article, the longitude and latitude chosen with a surface distance of 5 m are adopted as the axis unit for detail-level expression. As a consequence, the longitude resolution R_{lon} and latitude resolution R_{lat} of the geospatial category probability maps are calculated as follows:

$$S_{\text{lon}} = \text{Rcos}^2(\text{La})\sin^2\left(\frac{\text{rad(lon1)} - \text{rad(lon2)}}{2}\right) \tag{6}$$

$$S_{\text{lat}} = 2\text{R}\arcsin\left(\sqrt{\sin^2\left(\frac{\text{rad(lat1)} - \text{rad(lat2)}}{2}\right)}\right) \tag{7}$$

$$R_{\text{lon}} = \frac{|\text{lon1} - \text{lon2}|}{S_{\text{lon}}} \cdot \lambda \tag{8}$$

$$R_{\text{lat}} = \frac{|\text{lat1} - \text{lat2}|}{S_{\text{lat}}} \cdot \lambda \tag{9}$$

where, R is the radius of the Earth, and (lon1, lat1, lon2, lat2) is the latitude and longitude range of the airport. La is the center latitude of the airport. S_{lon} and S_{lat} are the surface distance along the longitude and latitude direction of the airport, respectively. λ is the map resolution, and the default setting is 5 m.

After defining the coordinate system and resolution of the map, the 2-D GD is used to model the category probability distribution of aircraft. Although alternative methodologies for modeling geospatial probability maps are available, GDs present unique advantages in addressing complex spatial relationships. On one hand, the well-understood properties of GD make it highly suitable for probability modeling. On the other hand, given the inherent positioning bias in SAR images, GD's smooth and continuous probability distribution helps model aircraft distribution information effectively. Specifically, the center longitude and latitude of the ground-truth box in the training set are the reference coordinates. Then, the category probability distribution of aircraft is determined by calculating the deviation between the provided latitude and longitude coordinates and the reference coordinates, as illustrated in Fig. 9. The formula is presented below

$$P(C_i|X, G) = \frac{1}{\sqrt{2\pi}\delta} e^{-\frac{1}{2\delta}\left(\left(\frac{G_x - \text{Box}_{\text{ctrlon}}}{R_{\text{lon}}}\right)^2 + \left(\frac{G_y - \text{Box}_{\text{ctrlat}}}{R_{\text{lat}}}\right)^2\right)} \tag{10}$$

where $P(C_i|X, G)$ is the probability density function (pdf) of category C_i on location G. (G_x, G_y) is the geographic coordinates (i.e., longitude and latitude) corresponding to position (x, y) on the input image. $(\text{Box}_{\text{ctrlon}}, \text{Box}_{\text{ctrlat}})$ represent the center longitude and latitude of the ground-truth box. δ is the standard deviation.

On this basis, as the historical images are progressively traversed, GDs are superimposed according to the category and position of the aircraft. Ultimately, the normalized geospatial category probability maps with multiple GDs are formed, in which the parking position of the aircraft will have a greater probability. In this way, the geospatial category relationships

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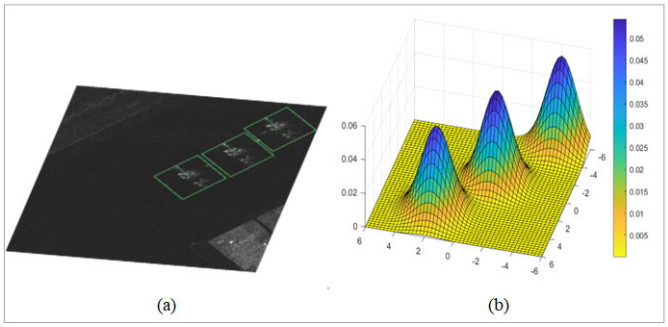


Fig. 9. Category probability map for the aircraft based on GD. (a) Slice image containing three aircraft and (b) 3-D diagram illustrating the position and the category probability distribution of aircraft using GD.

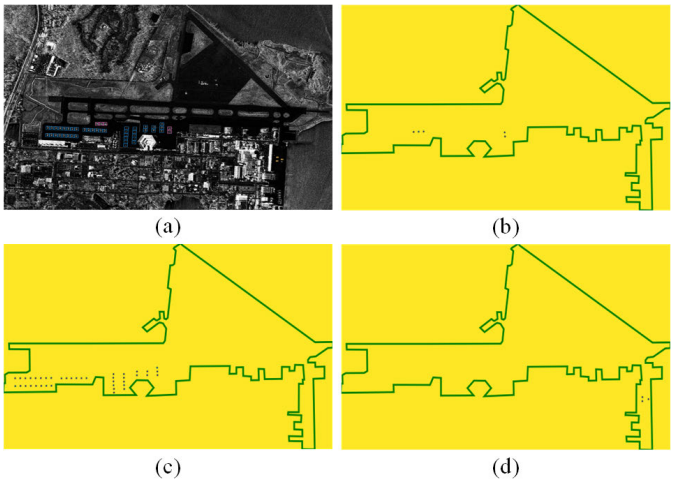


Fig. 10. Visualization of the geospatial category probability maps at the airport (I). (a) SAR image of the airport (I), which covers three types of aircraft. (b)–(d) Geospatial category probability maps corresponding to three different types of aircraft, respectively. The category probability map is composed of a series of grids, each of which depicts the position probability of the aircraft. The colors closer to yellow indicate a lower probability, while colors closer to blue indicate a higher probability.

between various types of aircraft in the historical images are comprehensively summarized. Fig. 10 presents the geospatial category probability maps of aircraft at the airport (I). In Fig. 10, each grid on the map represents the composite result of multiple GDs associated with its adjacent targets in multiple historical images.

2) Postprocess Model: After obtaining the geospatial category relationships, the posterior probability from the classifier of the oriented fine-grained aircraft detection network is modulated with the geographic category prior probability to enable more accuracy and stability of classification prediction. Let the input image be X, the output probability distribution $P(C_i|X,G)$ of the postprocess model can be expressed as follows:

$$P(C_i|X,G) = \min\left\{\left[P_{\text{posterior}}(C_i|X,G)+\beta * \sum_j {P^j}_{\text{prior}}(C_i|X,G)\right], 1\right\} \tag{11}$$

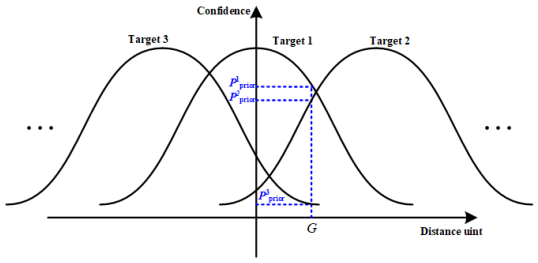


Fig. 11. Schematic of the prior probability distribution of category C_i on the geospatial category probability map. The prior probability of category C_i at geographic position G is determined by the composite result of multiple GDs associated with its adjacent targets across multiple historical images.

where C_i denotes the ith category. β is the weighting factor. $P_{\text{posterior}}(C_i|X,G)$ is the posterior probability from the classifier of the aircraft detection network. $\sum_j {P^j}_{\text{prior}}(C_i|X,G)$ denotes the total prior probability for category C_i at geolocation G on the geospatial category probability map, as illustrated in Fig. 11.

III. Experiments

In this section, experiments are conducted to confirm the effectiveness of our proposed network. First, we describe the experimental dataset, implementation details, and evaluation metrics. Then, the comparative analysis with state-of-the-art methods is presented. Finally, several ablation experiments are conducted to validate our design decisions.

A. Dataset Description

To evaluate the effectiveness of the proposed algorithm, a series of experiments are conducted on a self-constructed rotated SAR aircraft detection dataset (named SAR-RADD). Table I provides the fundamental information about the SAR-RADD. The SAR-RADD comprises 61 single-polarization panoramic SAR images with 1-m resolution, working in the C-band and spotlight mode. These satellite images cover the 16 airports with different periods. The aircraft in the dataset have been meticulously annotated and verified by skilled SAR image interpreters using OBBs. The aircraft are classified into 13 specific categories according to airport scenes, as shown in Fig. 12. Given the notable intraclass variations and subtle interclass distinctions, fine-grained aircraft detection based on SAR imagery poses a significant challenge. Meanwhile, we are committed to considering the possibility of sharing this dataset with the community in the future when the circumstances are more favorable. In our experiments, 17 SAR images from different periods are selected for independent testing, while the remaining 44 images are allocated for training. Examples from the SAR-RADD training set are presented in Fig. 13. Furthermore, Fig. 14 illustrates the number and scale distribution of different aircraft types in the training set, highlighting the class imbalances and the scale heterogeneity. The dataset will be available to the community in the future.

Moreover, we construct a geodatabase comprising two main components: geographic polygons and geospatial category probability maps for airports. Specifically, aircraft detection is a special case since aircraft are often parked at airports. The

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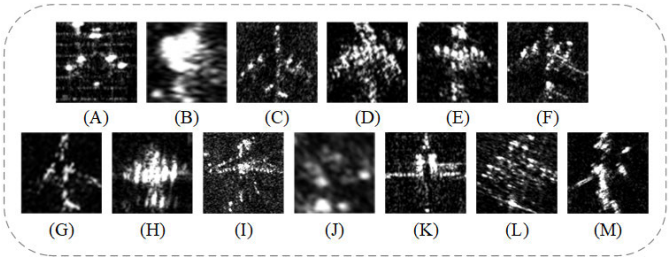


Fig. 12. Examples of 13 aircraft categories in SAR-RADD. We number the different categories of aircraft by the letters A–M.

TABLE I
FUNDAMENTAL INFORMATION ABOUT SAR-RADD

No.	Parameters	Description
1	Source	HiSea-1, Chaohu-1, Gaofen-3
2	Resolution	1m
3	Polarization	VV
4	Image mode	Spotlight
5	Radar frequency	C-band (5.4GHz)
6	Coverage	16 airport area
7	Annotation	Oriented bounding boxes (OBBs)
8	Category	13 categories
9	Image format	GeoTIFF[1]
10	Total Images	61 panoramic SAR images

[1] The SAR imagery products with GeoTIFF format contain geographic information.

geolocation of the airport can be determined in advance. Based on this knowledge, we manually create geographic polygons that delineate the contours of airports on open-source satellite optical images. This task is carried out on the well-established image processing platform, BigeMap, which can create polygons at any location on the map and supports the export of latitude and longitude boundaries of these polygons in KML format. The polygons are then stored in the database as geolocation priors. Furthermore, we use geographic information in collaboration with GDs to model the geospatial category relationships based on historical images in probability maps.

B. Implementation Details

The experiments were conducted on a Windows system with a single 24-GB NVIDIA 3090 GPU. Detailed environmental parameters are provided in Table II. Large-scale SAR images were adaptively cropped into 1024 × 1024 slices for model training. The optimization strategy used was AdamW with a weight decay of 0.05, using cosine annealing for the learning rate schedule, starting with an initial learning rate of 0.0005. To ensure a fair comparison, all the networks were trained on the same dataset for 100 epochs and saved the optimal parameter model. Furthermore, the inference was carried out on the same testing data with a batch size of 1 to record evaluation results.

C. Evaluation Metrics

In our article, two evaluation metrics are adopted to quantitatively evaluate the fine-grained aircraft detection performance of the algorithm: mean average precision (mAP) and frames per second (FPS). On one hand, FPS represents the number of images processed per second, providing insight into the real-time processing efficiency of the network. On the other hand, mAP is the mean of average precision (AP) across all the categories, which is a widely common metric for an in-depth and comprehensive evaluation of detection and classification performance. AP is calculated by combining precision (P) and recall (R). Among them, the confusion matrix is essential for calculating P and R, which are also important metrics for assessing the classification quality of the model. P assesses the proportion of true positives (correct predictions) against the total positive predictions, while R measures the percentage of true positives in labeled instances. The relevant calculation formula can be expressed as

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{12}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{13}$$

$$\mathrm{AP} = \int_0^1 P(R) dR \tag{14}$$

$$\mathrm{mAP} = \frac{\sum_{i=1}^{N_c} AP_i}{N_c} \tag{15}$$

where TP, FP, and FN represent true positives, false positives, and false negatives. N_c is the number of categories.

D. Comparison to State-of-the-Arts

1) Overall Performance Comparison: In this section, a series of comparative experiments are carried out. The proposed method is compared with five state-of-the-art methods, i.e., the classical two-stage detector (RoI Transformer [33], Oriented R-CNN [34]) and the single-stage detector (RetinaNet-OBB [35], Oriented RepPoints [36], RTMDet [27]). For fairness, the control variable principle is adopted and the proposed geospatial-driven AD algorithm is used as the preprocessing. On this basis, the performance of fine-grained aircraft detection is compared between the proposed method and mainstream methods, as detailed in Table III.

According to Table III, the Oriented R-CNN, a two-stage detector, has exhibited robust performance in fine-grained aircraft detection in SAR imagery, achieving an mAP of 69.2%. In contrast, single-stage algorithms such as RetinaNet-OBB and Oriented RepPoints demonstrate efficiency advantages over two-stage detectors but display weaker performance in both detection and classification tasks. RTMDet, a high-precision, low-latency, and single-stage detector, is developed by integrating insights from the YOLO series and other advanced detectors. Therefore, RTMDet performs well in mainstream algorithms, with an mAP of 74.7%. Compared with mainstream algorithms, our proposed method is competitive, with an mAP of 81.2%. From the perspective of the efficiency of the algorithm, our algorithm has good timeliness, and the inference speed can reach 42 frames/s.

Fig. 15 shows the visualization results of the proposed method on the SAR image of the airport (II). There are only five misclassifications (highlighted by green ellipses), two missed detections (highlighted by yellow ellipses), and one false alarm that occurred near the airport boundary (highlighted by a red ellipse). This indicates satisfactory

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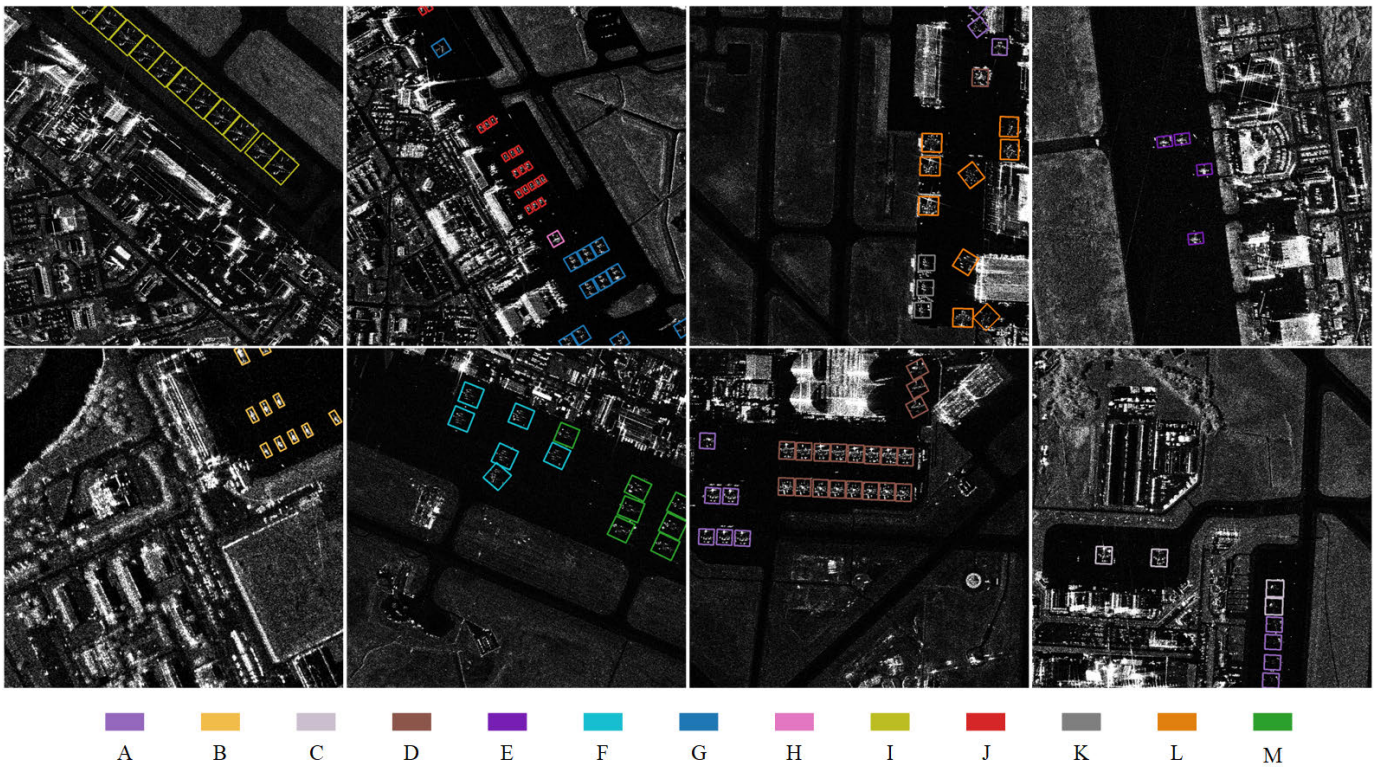


Fig. 13. Examples of aircraft in SAR-RADD consist of 13 specific categories, with each aircraft annotated using OBBs.

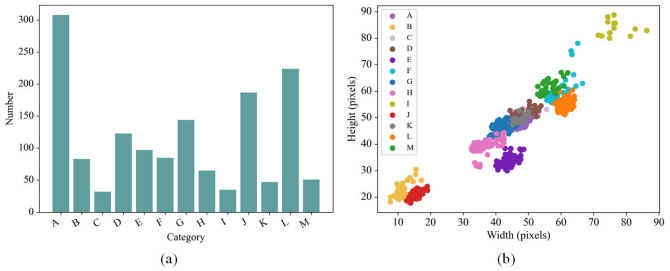


Fig. 14. Illustration of the number and scale distribution of various types of aircraft in the SAR-RADD training set. (a) and (b) Number and size distributions of aircraft in the SAR-RADD training set, respectively. All the targets undergo meticulous manual annotation by proficient SAR image interpreters. The aspect ratio of the bounding box reveals an approximately linear correlation, revealing notable scale heterogeneity arising from unique scale attributes among categories.

TABLE II
IMPLEMENTATION DETAILS OF ENVIRONMENTAL PARAMETERS

Name	Experimental environment
Operating system	Windows 10
CPU	Intel Core i9-12900 CPU @3.19GHz
Memory	24GB
Programming language	Python 3.9
Programming framework	Pytorch 1.11.0

performance and validates the effectiveness of the proposed method.

2) Confusion Matrix Analysis: Confusion matrix is used as a visual tool to provide insights into the accuracy and misclassifications of algorithms. Fig. 16 reports the confusion matrixes generated by the proposed method and mainstream object detection networks.

TABLE III
COMPARISON TO STATE-OF-THE-ART METHODS

Method	Backbone	Input size	mAP (%)	FPS
RoI Transformer	ResNet50	1024×1024	68.5	17.5
Oriented R-CNN	ResNet50	1024×1024	69.2	14.2
RetinaNet-OBB	ResNet50	1024×1024	65.1	26.3
Oriented RepPoints	ResNet50	1024×1024	59.4	23.6
RTMDet	CSPNetXt	1024×1024	74.7	46.2
Ours	CSPNetXt	1024×1024	81.2	42.0

From binary classification, as shown in the confusion matrix's final column, the two-stage algorithms outperform the single-stage algorithms in accurately distinguishing aircraft from the surrounding background. When it comes to fine-grained aircraft classification, our algorithm particularly focuses on the confusion matrix's diagonal, emphasizing more precise aircraft classification. Notably, the small size of helicopters (aircraft B) and fighters (aircraft J) makes it difficult to capture effective discriminative features as the network layers increase, which in turn complicates their differentiation from background or similar targets. It can be observed that confusion between helicopters and fighters is a significant challenge across the most mainstream methods. This challenge is effectively addressed by RTMDet and our proposed method. Owing to the well-designed RSIM and relational prior-based reasoning strategy, our method exhibits the lowest missed detections and the highest classification accuracy for helicopters and fighters.

Furthermore, there is significant mutual confusion between aircraft E and H across various comparison methods. Examination of failure instances revealed a notable similarity between aircraft E and H, after SAR imaging,

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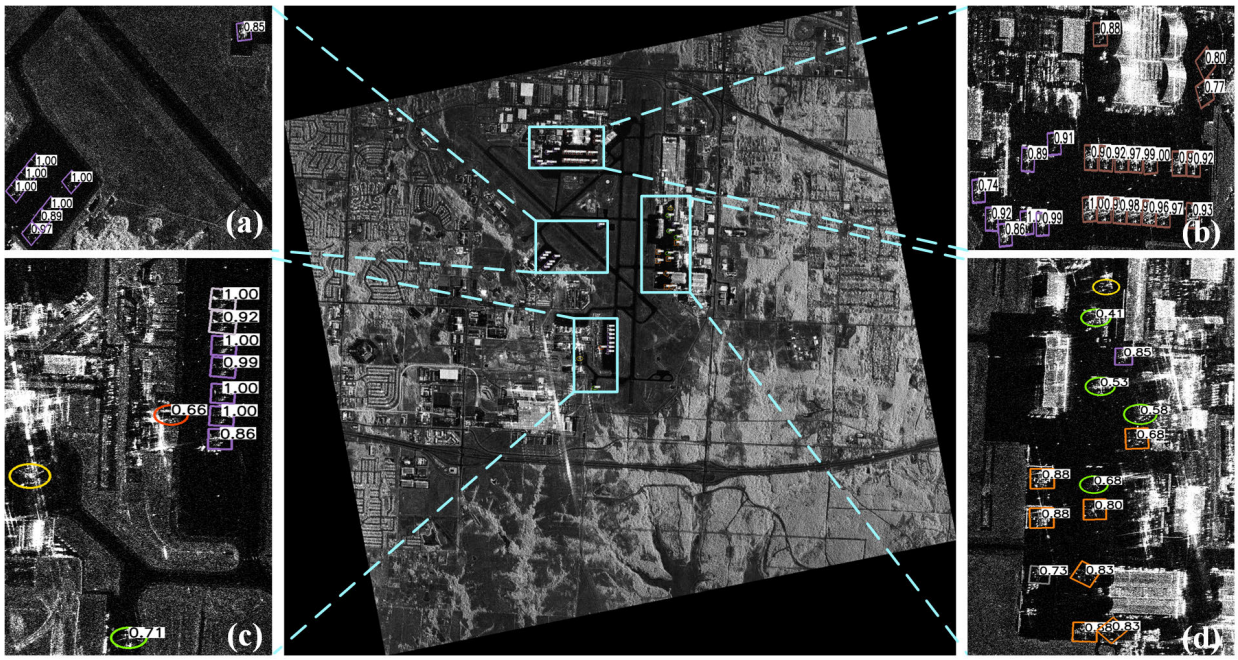


Fig. 15. Visualization results of the proposed method on the SAR image of the airport (II). For clarity, the rectangle boxes with different colors represent different categories of aircraft that have been correctly detected and classified. The misclassification, missed detection, and false detection are represented by green ellipses, yellow ellipses, and red ellipses, respectively. (a)–(d) in Fig. 15 correspond to detailed views of the regions of interest within the overall scene image.

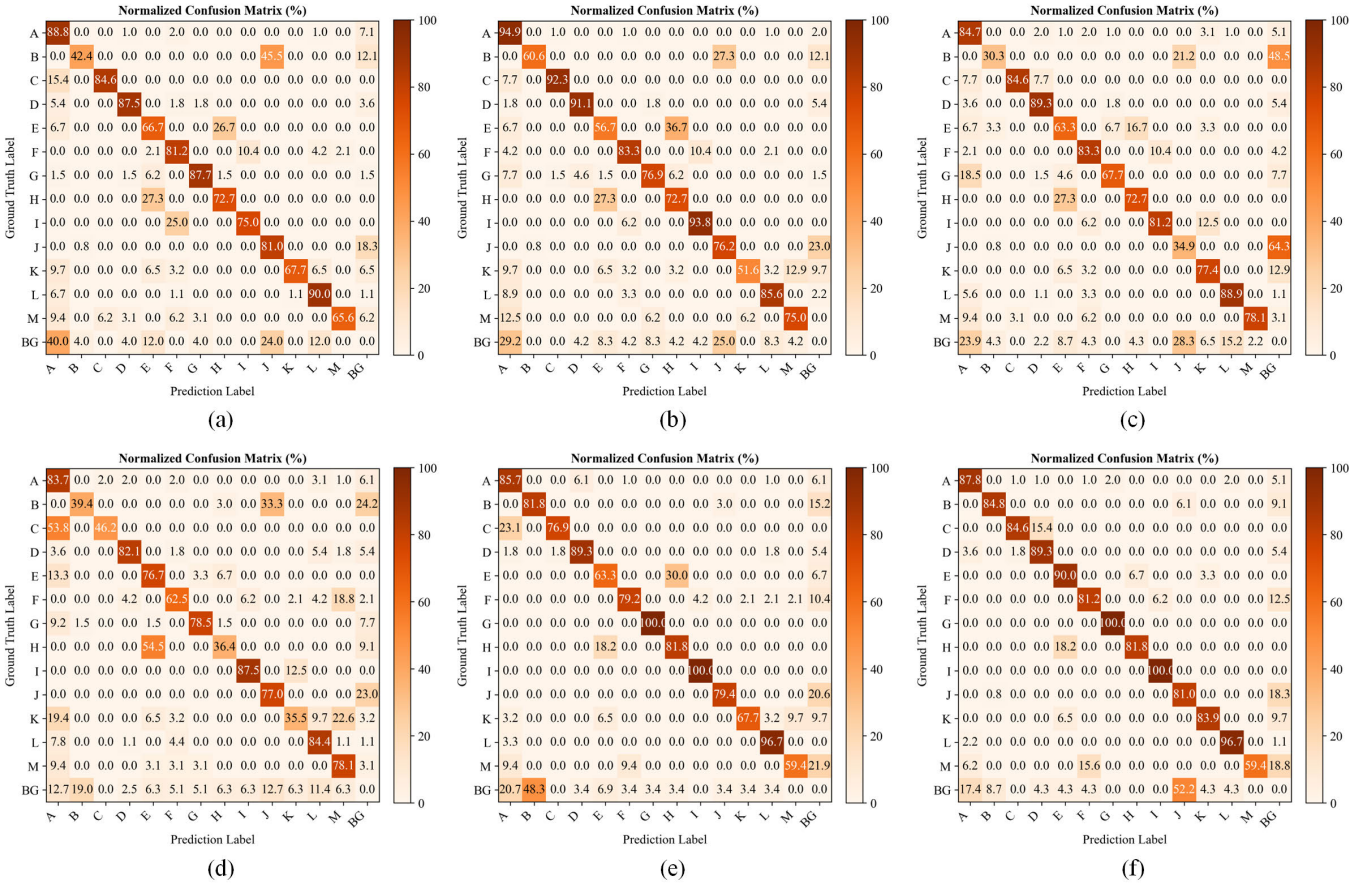


Fig. 16. Normalized confusion matrix of different networks. (a)–(f) RoI Transformer, Oriented R-CNN, RetinaNet-OBB, Oriented RepPoints, RTMDet, and our proposed method, respectively. In the confusion matrix, A–M is the different category of aircraft, and BG is the background.

leading to confusion within the feature space, as shown in Fig. 17. In contrast, our method attains superior classification performance through meticulous feature engineering and relational prior-based reasoning, yielding a classification accuracy of 90% for aircraft E and 81.8% for aircraft H.

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TABLE IV
EFFECTS OF COMPONENTS IN GEOSPATIAL-DRIVEN AD

AD with horizontal box	AD with geographic polygon	RIFT	mAP (%)	Avg. time (s)
–	–	–	61.1	11.27
✓	–	–	68.7	5.61
–	✓	–	65.1	5.62
✓	–	✓	69.4	8.26
–	✓	✓	74.7	8.29

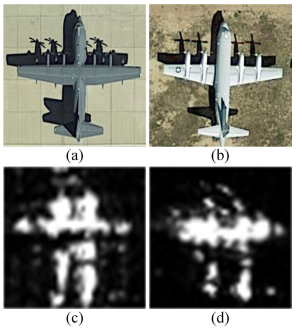


Fig. 17. Visualization of instances with misclassification. (a) and (b) Optical images of the aircraft E and H, respectively, and the corresponding SAR images are shown in (c) and (d).

E. Ablation Experiments

In this section, we investigate the effects of the proposed components within our framework and explore the effect of hyperparameters on relational prior-based reasoning. Through systematic and rigorous experiments and analyses, the goal is to acquire a comprehensive understanding of the factors influencing the effectiveness of the proposed framework in SAR imagery and to offer valuable guidance for optimizing performance.

1) Effects of the Component in Geospatial-Driven AD: To assess the impact of individual components within the proposed geospatial-driven AD when applied to large-scene SAR images, RTMDet is used for fine-grained aircraft detection. Subsequently, ablation experiments are conducted, and the results are presented in Table IV. As shown in Table IV, when large-scale SAR images are directly detected by sliding windows, the detection accuracy and efficiency of the algorithm are poor. The average test time is up to 11.27 s. After using AD, the detection range and background interference are reduced. It can be observed that both the average test time and mAP values have greatly improved. The average testing time is approximately 5.6 s. However, due to the inherent positioning errors in the imaging process, AD based on geographic information matching is inaccurate, which affects the overall performance of fine-grained aircraft detection. After using the RIFT algorithm to rectify geographic information bias, a notable performance enhancement can be observed with both the horizontal boxes (used in [7]) and the geographic polygons proposed in this article. In comparison, the geographic polygon aligns more closely with the airport's geometry, thereby mitigating background clutter interference and facilitating optimal detection performance. Consequently, our proposed method exhibits satisfactory detection efficiency and performance, attaining an mAP of 74.7% with an average testing time of 8.29 s.

TABLE V
EFFECTS OF RSIM AND RELATIONAL PRIOR-BASED REASONING

RSIM	Relational Prior-based Reasoning	mAP	Avg. time (s)
–	–	74.7	8.29
✓	–	79.1	8.43
–	✓	79.5	8.42
✓	✓	81.2	8.58

2) Effects of RSIM and Relational Prior-Based Reasoning: The effectiveness of the RSIM and relational prior-based reasoning in the proposed framework is validated through ablation experiments, with comprehensive details presented in Table V. The RTMDet is used as the baseline. It can be seen that the incorporation of the RSIM module produced better results than the baseline, with a gain of 4.4% for mAP. This phenomenon highlights the remarkable discriminative capability of the RSIM. From another view, with relational prior-based reasoning, at 4.8%, a gain of mAP is obtained. From an overall point of view, the combined use of RSIM and relational prior-based reasoning attained an mAP of 81.2%, nearly 6.5% higher than the baseline. Meanwhile, the decrease in average time is not considerable, indicating that the performance gains from the proposed method are achieved without significant compromise on operational efficiency.

Fig. 18 illustrates the visualization of the results in the ablation experimental setup. Compared with the ground truth [as shown in Fig. 18(b)], the RTMDet baseline [as shown in Fig. 18(c)] presents obvious false alarms and omissions. It can be seen that only two aircraft are accurately detected and classified, three aircraft are misclassified (green ellipses), six aircraft are missed detection (yellow ellipses), and three background targets are erroneously identified as helicopters (red ellipses). However, with the addition of the RSIM module (as shown in Fig. 18(d), the false alarms are eliminated, and omissions are reduced. Furthermore, both the comparisons between Fig. 18(c) and (e), or (d) and (f) reveal that relational prior-based reasoning can improve the model's classification performance. This phenomenon highlights the benefits of integrating geospatial category knowledge, facilitating more precise category predictions. With the advantages of relational prior-based reasoning and RSIM, the proposed method [as shown in Fig. 18(f)] excels in detecting and distinguishing subtle distinctions among classes. As a result, the classification results align more closely with the ground truth, showcasing satisfactory performance.

3) Effects of Hyperparameters on the Performance of Relational Prior-Based Reasoning: In this section, we analyze the impact of different hyperparameters in relational prior-based reasoning, as shown in Fig. 19. RTMDet-RSIM is the baseline. Following the incorporation of relational prior-based reasoning, a notable enhancement in mAP is observed

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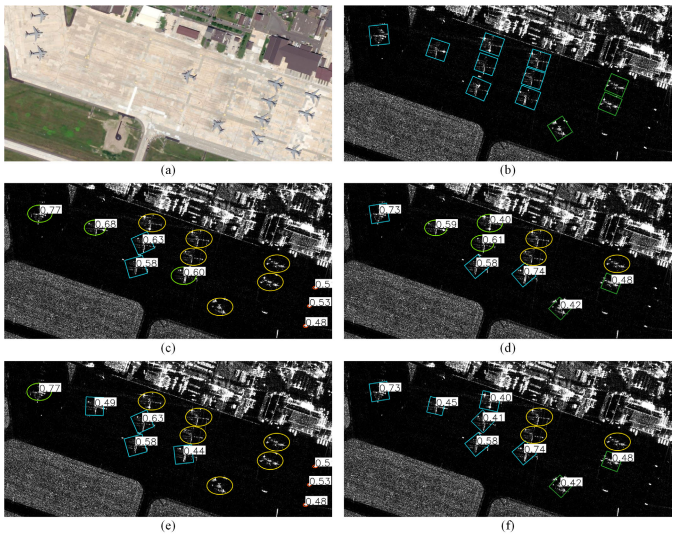


Fig. 18. Visualization results of ablation experiments. (a)–(f) Optical image, ground truth, RTMDet baseline, RTMDet w/RSIM, RTMDet w/relational prior-based reasoning, and our proposed method. Rectangle boxes of different colors represent aircraft of various categories. For clarity, the rectangle boxes with different colors represent different categories of aircraft that have been correctly detected and classified. The misclassification, missed detection, and false detection are represented by green ellipses, yellow ellipses, and red ellipses, respectively.

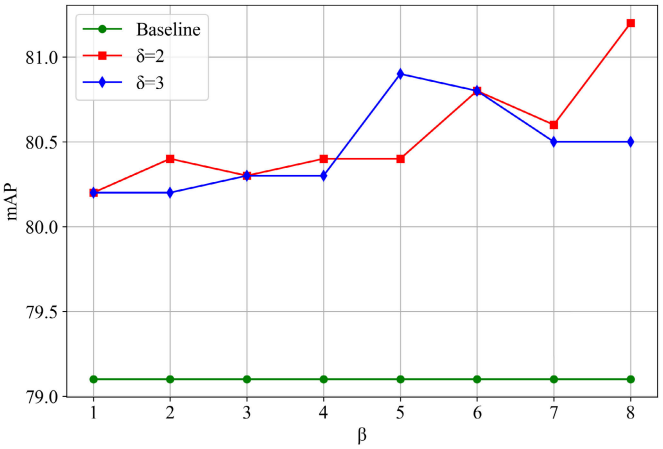


Fig. 19. Effects of hyperparameters on the performance of relational prior-based reasoning. β is the weighting factor in relational prior-based reasoning and δ is the standard deviation of GD.

relative to the baseline. Specifically, the optimal performance is attained when parameters $\delta = 2$ and $\beta = 8$, yielding an optimal mAP of 81.2%. This phenomenon proves that relational prior-based reasoning can effectively mitigate CNN's category prediction bias. Furthermore, the alternative ranges of δ and β are practical and moderate as well as provide a degree of relaxation in the location distribution of potential objects, thereby enhancing the framework's utility in practical applications.

IV. Conclusion

In this article, a novel geospatial contextual prior-enabled knowledge reasoning framework for fine-grained aircraft detection in SAR images is proposed. The framework takes advantage of geolocation priors, scattering characteristics of aircraft, and geospatial category relationships. First, a geospatial-driven AD method uses geolocation priors for rapid AD. Then, an oriented fine-grained aircraft detection network, RTMDet-RSIM, produces advanced representations of aircraft in various orientations and sizes, thereby improving the performance of fine-grained aircraft detection. Finally, a relational prior-based reasoning strategy is proposed to improve classification performance by modeling geospatial category relationships from historical scenes. Experiments demonstrate the superiority of the proposed method and its potential for practical application.

However, aircraft with small sizes and plenty of intra-class variations remain challenging in our proposed framework. In the future, we will focus on efficiently capturing fine-grained attributes of aircraft using spatial templates or super-resolution modules to improve the performance of fine-grained aircraft detection in panoramic SAR images.

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